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Project Title: DATA ANALYSIS USING POWER BI

Submitted in partial fulfillment of the requirements for the award of the
degree of

Master of Science (Data Science) MSc (DS)

By

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Enrollment No: 024MSD110062

Under the guidance of

Guide/Mentor Name: Mr. Sudhanshu Narain

Certificate

Copy to the certificate received from Qollabb to be pasted here.

Declaration

I, **ASIF ALI BHATI**, hereby solemnly declare that the project report titled "**DATA ANALYSIS USING POWER BI**" submitted in partial fulfilment of the requirements for the award of the degree of **Master of Science (Data Science) MSc (DS)** is my original work.

I further declare that:

- This project has been carried out by me during the academic year 2026 under the supervision of **Mr. Sudhanshu Narain**.
- The work has not been submitted previously to any other university, institution, or examination body for the award of any degree, diploma, or certification.
- All sources of information used in this report have been duly acknowledged and referenced in accordance with academic ethics and plagiarism norms.
- The data presented in this report is authentic to the best of my knowledge and has not been fabricated or manipulated.
- I understand that if any part of this declaration is found to be false, my project may be rejected and disciplinary action may be taken as per institutional rules.

Place: Sikar, Rajasthan

Date: 21-June-26

Student Signature: _____

Student Name: ASIF ALI BHATI

Enrollment No: O24MSD110062

Acknowledgement

I would like to express my sincere gratitude to my esteemed mentor, Mr. Sudhanshu Narain, whose invaluable guidance, continuous encouragement, and profound insights were instrumental in shaping the success of this major project. I am also deeply thankful to the dedicated faculty members of Chandigarh University for providing a rigorous academic foundation and a highly supportive learning environment throughout my Master of Science in Data Science program. Furthermore, my heartfelt thanks go to the management and team at Riyadh Art for allowing me to work on this prestigious mega-project, granting access to essential resources, and fostering a collaborative environment that made the completion of this visual analytics dashboard possible.

Project Disclaimer

THE DATA UTILIZED IN THIS PROJECT IS SAMPLE DATA AND IS INTENDED SOLELY FOR THE FULFILLMENT OF THE ACADEMIC REQUIREMENTS OF THE MSC DATA SCIENCE PROGRAM. THIS DATA IS THE PROPERTY OF THE SAUDI ARABIAN GOVERNMENT. THE UNAUTHORIZED USE, REPRODUCTION, OR DISTRIBUTION OF THIS DATA AND THE ASSOCIATED DASHBOARD FOR ANY PURPOSE OUTSIDE OF THIS ACADEMIC PROJECT IS STRICTLY PROHIBITED. ALL INSIGHTS AND VISUALIZATIONS ARE FOR EDUCATIONAL PURPOSES ONLY.

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Chapter 1: Introduction

1.1 Background of the Project

Riyadh Art is one of the four mega projects in Riyadh, Saudi Arabia, representing a major cultural and architectural milestone. The project consists of 13 sub-projects distributed across the city, involving a global pool of artists tasked with creating city-wide art installations. Managing the massive volume of data associated with these 13 sub-projects, including artist vetting, program participation, and artwork delivery, is a significant administrative challenge.

1.2 Problem Statement

Previously, Riyadh Art relied on a centralized Excel file to manage its artist database. This system proved highly inefficient as the number of participating artists and artworks grew. Stakeholders faced 'Data Blindness,' where it was nearly impossible to extract high-level trends or monitor real-time progress across multiple programs. There was an urgent need for an interactive, automated BI solution to replace manual tracking.

1.3 Objectives of Study

- Design a Star Schema data model to link Artist and Artwork databases.
- Implement a Power BI dashboard with 11 interactive charts and 2 KPI multicards.
- Automate status tracking (Awarded, Qualified, etc.) using complex DAX measures.

Chapter 2: System Study

2.1 Existing System Analysis

The existing system was based on static spreadsheets. Limitations included data silos, high risk of manual entry errors, and lack of visual storytelling. Stakeholders could not easily identify bottlenecks in artwork delivery or demographic trends.

2.2 Proposed System

The proposed system uses Microsoft Power BI to provide a 360-degree view of the database. It features dynamic filtering, automated KPI calculations, and geographic mapping of artists, ensuring data-driven decision-making for the 13 sub-projects.

Chapter 3: System Analysis

3.1 Requirement Gathering

Functional requirements included tracking 681 unique artists and 765 artworks. Users needed to filter data by Program, Nationality, Gender and Vetting Status.

3.2 Feasibility Study

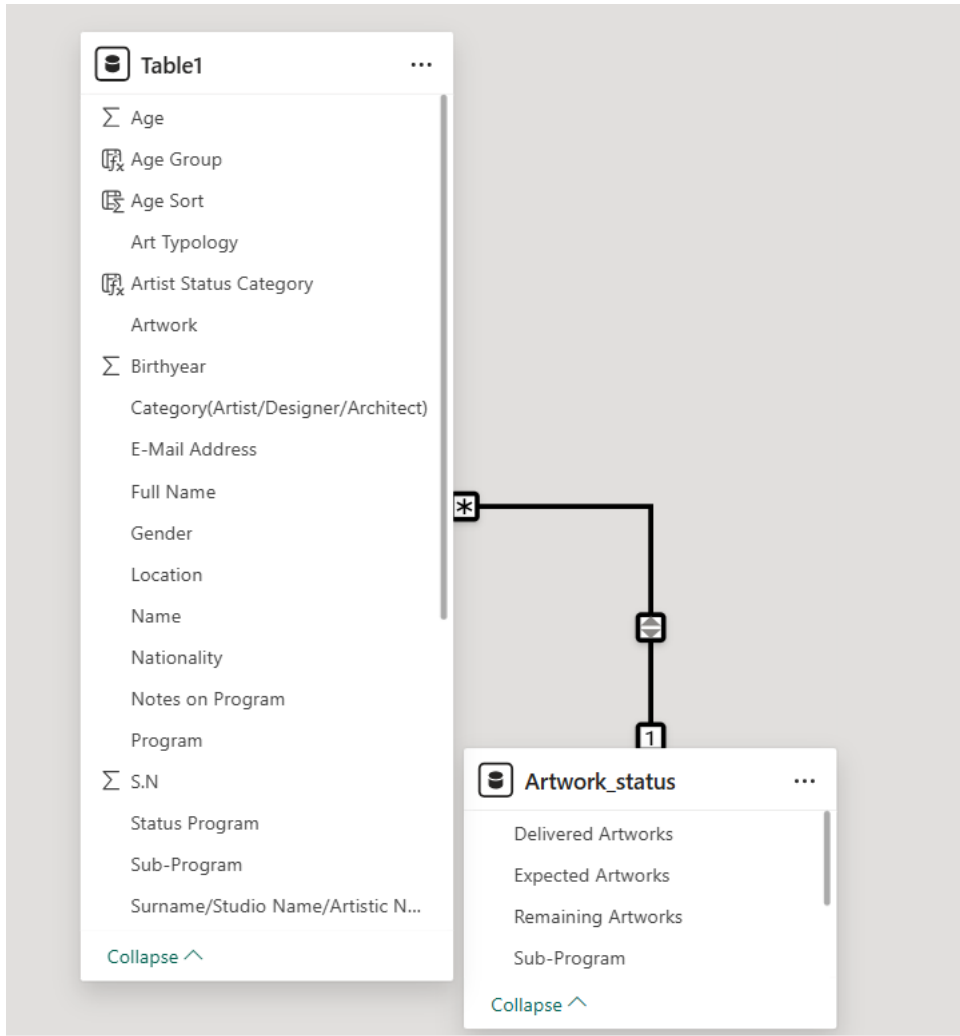
Technical Feasibility: Power BI is highly capable of handling the Riyadh Art datasets.

Operational Feasibility: The team can easily navigate the dashboard using intuitive slicers and cross-filtering.

Chapter 4: System Design

4.1 Entity Relationship Diagram (ERD)

The model follows a Star Schema with a one-to-many relationship linking Table1 (Artist Fact) and Artwork_status (Dimension).



From table

Table1

Program	S.N	Status Program	Sub-Program	Surname/Stu...	Vetting Process	Website
Art on the Mo...	51	Not Qualified	Art on the M...	AlHashemi	YES	https://r
Art in Transit	63	Qualified	Art in Transit	AlMajhadi	YES	-
Art on the Mo...	65	Not Qualified	Art on the M...	AlMazrooa	null	http://ec

To table

Artwork_status

Delivered Art...	Expected Art...	Remaining Ar...	Sub-Program
null	0	0	Urban Art Lab
null	34	34	Joyous Gardens
11	25	14	Jewels in Riya...

Cardinality

Many to one (*:1)

Cross-filter direction

Both

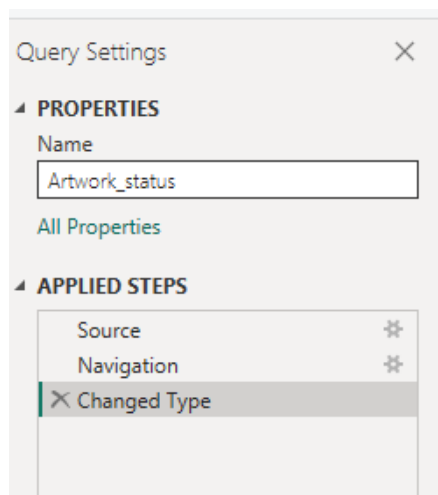
☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

4.2 Power Query ETL Pipeline

Steps included Source Navigation, Data Type conversion (SNR/Age).



fx = Table.TransformColumnTypes(Artwork_status_Table,({"Sub-Program", type text}, {"Expected Artworks", Int64.Type}, {"Delivered Artworks", Int64.Type}, {"Remaining Artworks", Int64.Type}))

Sub-Program	Expected Artworks	Delivered Artworks	Remaining Artworks
1 Urban Art Lab	0	null	0
2 Joyous Gardens	34	null	34
3 Jewels in Riyadh	25	11	14
4 West Entrance Gateway	0	null	0
5 Welcoming Gateways	1	null	1
6 Art on the Move	12	3	9
7 Art in Transit	107	null	107
8 Urban Flow	0	null	0
9 HiB9 - Illuminated Bridges	5	null	5
10 Hidden River - Art Trail	155	62	93
11 Riyadh Icon	0	null	0
12 Noor Riyadh	426	426	0
13 Tuwaiq International Sculpture Symposium	0	null	0

Query Settings

PROPERTIES

Name
Table1

All Properties

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Renamed Columns1**

fx = Table.RenameColumns(#"Changed Type",({"Status Program 1#(If)(Awarded/Qualified/Not Qualified/Not Awarded)", "Status Program"}))

S.N	Surname/Studio Name/Artistic Name	Name	Nationality	E-Mail Address	Category (Artist/Desi...	Website	Ge
1	1 Abbas	Hanira	Pakistan	null	Artist	https://www.hamraabbas.info/	F
2	2 Abboud	Michel	United States of America	null	Artist	www.michelabboud.com	M
3	3 Abdulhadi	Fatma	Saudi Arabia	null	Artist	https://fatmaabduhadi.com/	F
4	4 Abduljawad	Omar	Saudi Arabia	null	Artist	https://riyadhart.sa/en/artists/omar-abduljawad/	M
5	5 Abdulsteeef	Bayan	Saudi Arabia	null	Artist	https://www.lbehance.net/bayanabdusteeef	F
6	6 Abramović	Marina	Serbia	null	Artist	http://www.marinaabramovic.com/bio.html	F
7	7 Abu Abdallah	Sarah	Saudi Arabia	null	Artist	https://riyadhart.sa/en/artists/sarah-abu-abdallah/	F
8	8 Abu Hamdan	Lawrence	Jordan	null	Artist	http://lawrenceabuhamdan.com/	M

4.3 Custom Column Addition

Added 'Age Group' and 'Age Sort' to enable logical demographic clustering in clustered bar charts.

```

1 Age Group =
2 SWITCH(
3     TRUE(),
4     [Age] < 30, "20-29",
5     [Age] < 40, "30-39",
6     [Age] < 50, "40-49",
7     [Age] < 60, "50-59",
8     [Age] < 70, "60-69",
9     [Age] < 80, "70-79",
10    [Age] >= 80, "80+",
11    "Other"
12 )

```

```

1 Age Sort =
2 SWITCH(
3     TRUE(),
4     [Age] < 30, 1,
5     [Age] < 40, 2,
6     [Age] < 50, 3,
7     [Age] < 60, 4,
8     [Age] < 70, 5,
9     [Age] < 80, 6,
10    [Age] >= 80, 7,
11    8
12 )

```

Chapter 5: System Implementation

5.1 Dashboard Layout and Components

The dashboard is organized into logical sections for maximum executive impact:

5.1.1 KPI Multicard Headers

- Artist KPIs: Total Unique Artists, Awarded, Not Awarded, Qualified, Not Qualified, Repeated Artists, and Artist Repetition.
- Artwork KPIs: Expected, Delivered, and Remaining Artworks.

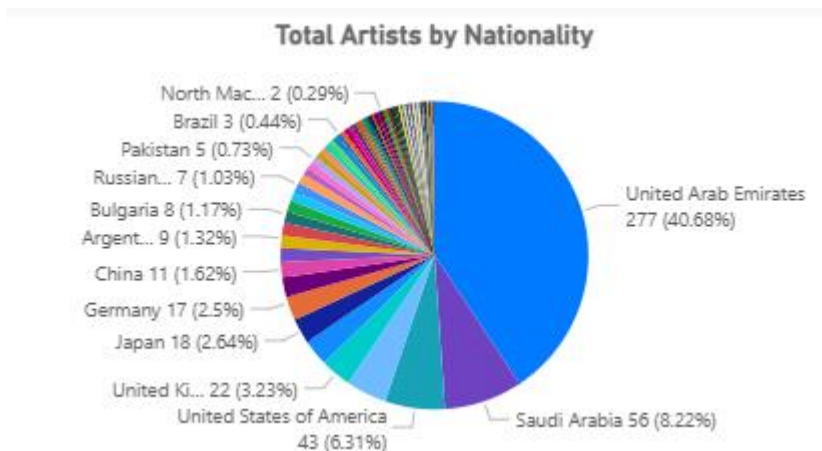
ARTIST's							ARTWORK's		
681	334	47	169	130	2	8	765	502	263
Total Unique ...	Awarded	Not Awarded	Qualified	Not Qualified	Repeated Artis...	Artist Repetition	Expected Artworks	Delivered Artworks	Remaining Artworks

5.1.2 Visualization Dictionary (The 11 Charts)

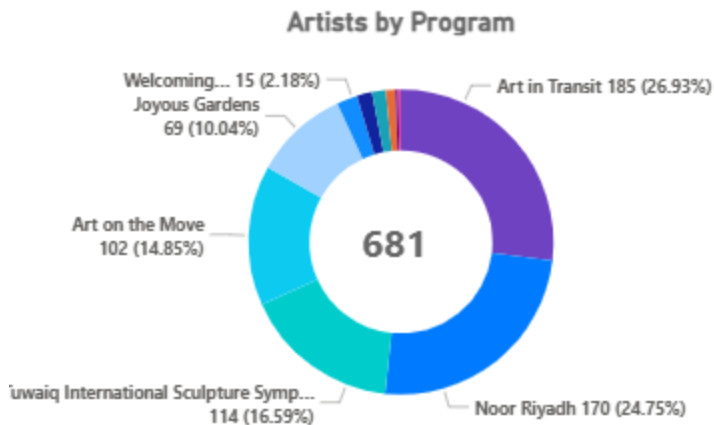
1. Azure Map (Artist by Nationality): Visualizes global footprints using blue markers.



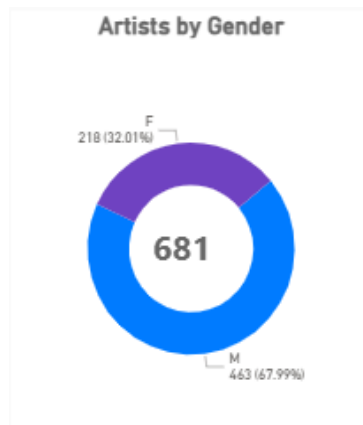
2. Pie Chart: Total artists by nationality using DistinctCount logic.



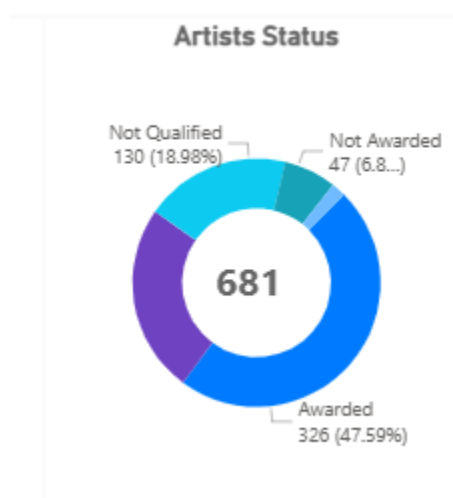
3. Donut Chart: Artists per program (tracking across 13 sub-projects).



4. Donut Chart: Gender distribution (Male vs Female).



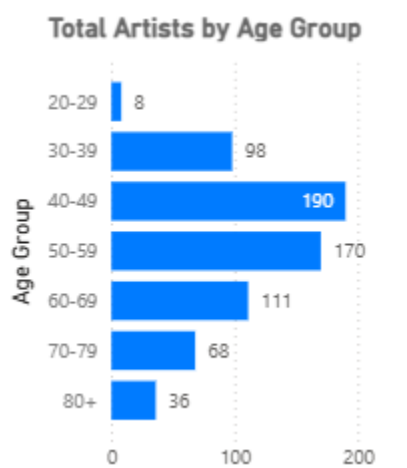
5. Donut Chart: Artist vetting status (Qualified, Approved, etc.).



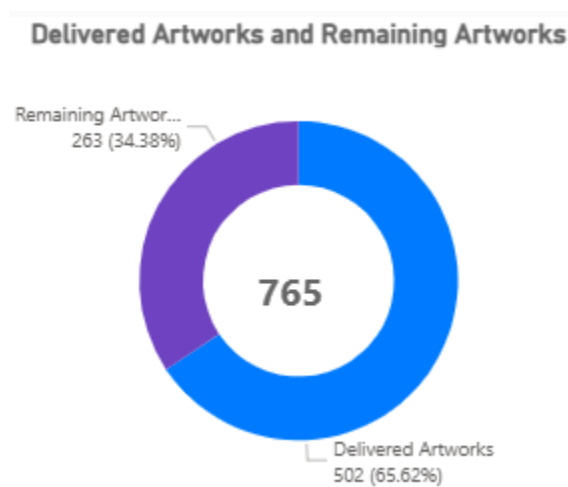
6. Clustered Bar Chart: Top 10 Nationalities using TopN visual filters.



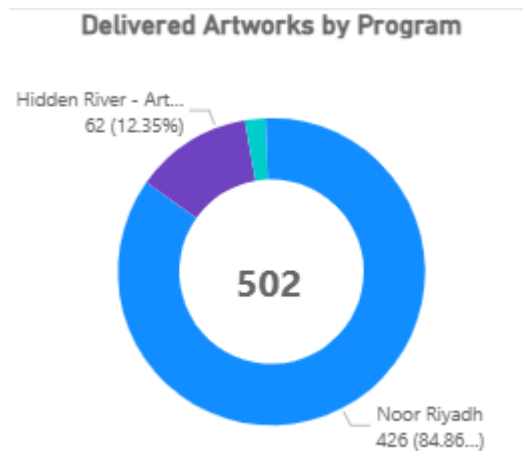
7. Clustered Bar Chart: Total artists by Age Group custom column.



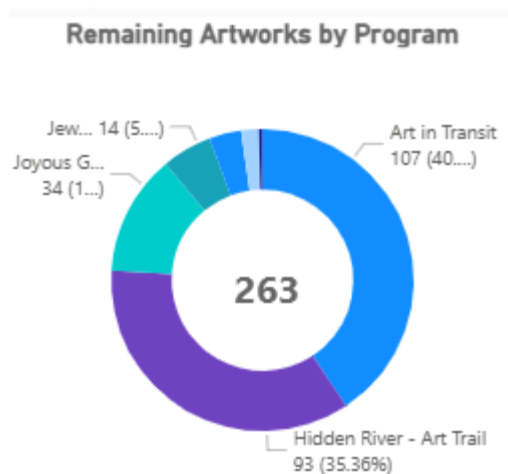
8. Donut Chart: Total delivered vs. Remaining artworks.



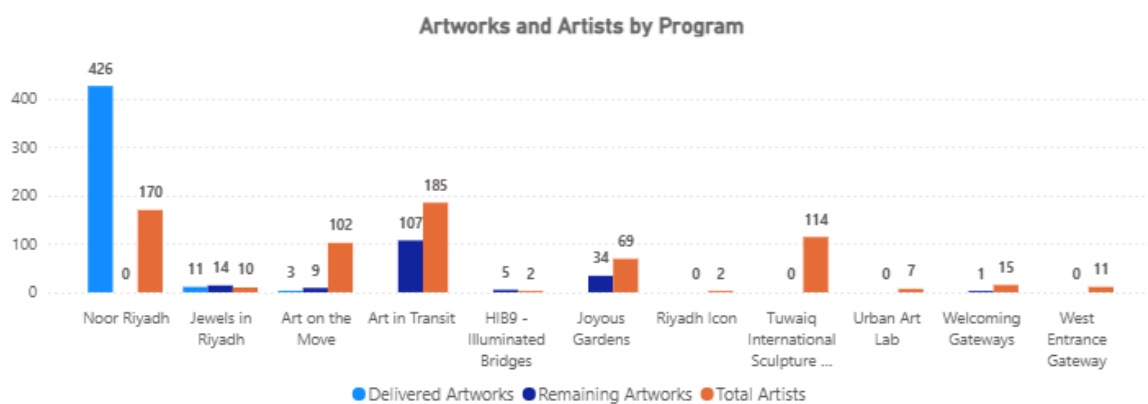
9. Donut Chart: Delivered artworks categorized by program.



10. Donut Chart: Remaining artworks categorized by program.



11. Clustered Column Chart: Comparative view of Artworks and Artists per program.



5.2 DAX Measure Repository

- Total Artists = DISTINCTCOUNT('Table1'[Alpha-2])
- Count Awarded = CALCULATE(COUNT('Table1'[Alpha-2]), 'Table1'[Status Program] = 'Awarded')
- Count Not Awarded = CALCULATE(COUNT('Table1'[Alpha-2]), 'Table1'[Status Program] = "Not Awarded")
- Count Qualified = CALCULATE(COUNT('Table1'[Alpha-2]), 'Table1'[Status Program] = "Qualified")
- Count Not Qualified = CALCULATE(COUNT('Table1'[Alpha-2]), 'Table1'[Status Program] = "Not Qualified")
- Artist Repetitions = MAXX(VALUES('Table1'[Alpha-2]), CALCULATE(COUNT('Table1'[Alpha-2])))
- Total Unique Artists (Global) = CALCULATE(DISTINCTCOUNT('Table1'[Alpha-2]), ALLSELECTED('Table1'))
- Count of Repeated Artists = Filter logic for multi-program participation.

Chapter 6: Testing

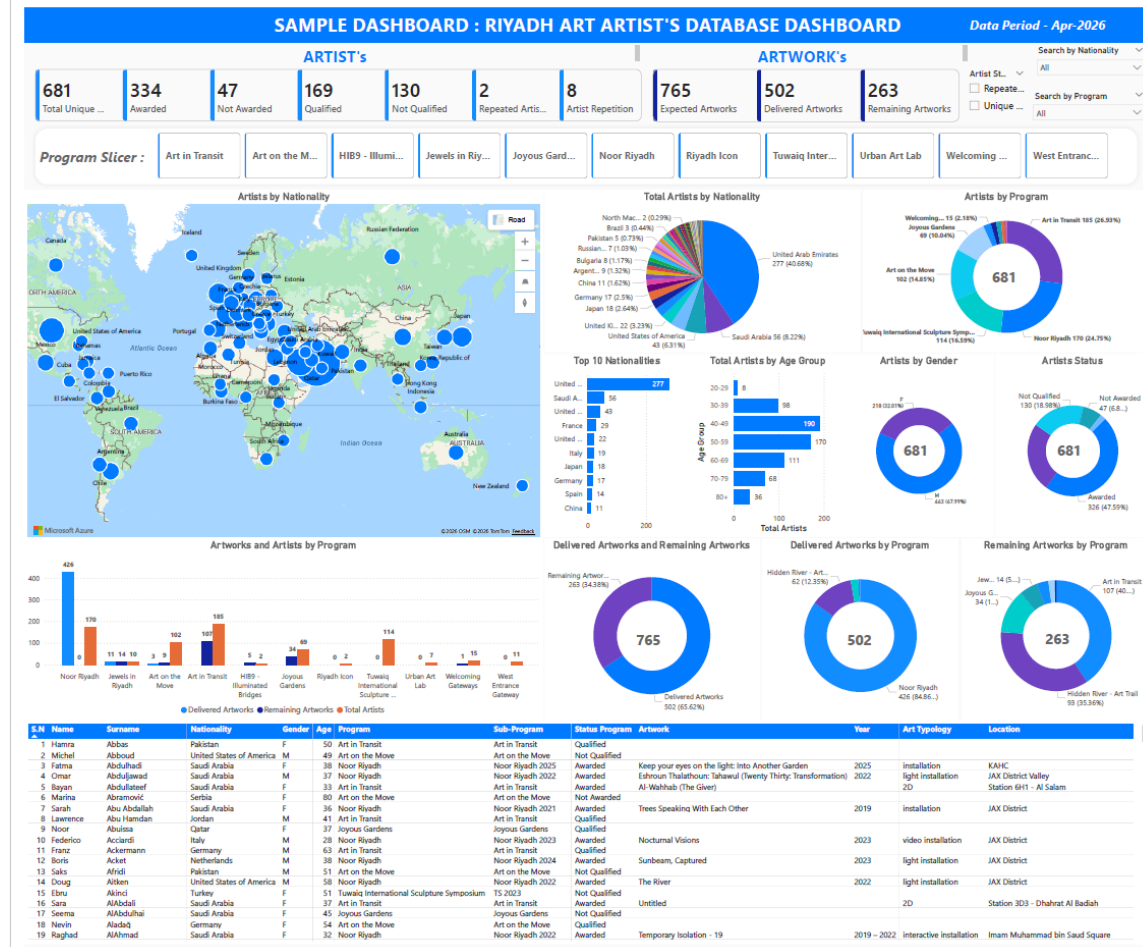
6.1 Data Validation Tests

Test Case 1: Filter dashboard by 'Art in Transit'. Result: Displays 185 Artists. (Passed)

Test Case 2: Verification of Global Unique Count. Result: 681. (Passed)

Chapter 7: Results & Discussion

The analysis reveals significant clusters in the 40-49 age demographic and strong delivery performance in the 'Noor Riyadh' program. The current delivery rate stands at 55.68% (426 of 765 artworks).



Chapter 8: Conclusion

The dashboard provides a scalable, automated solution for Riyadh Art, replacing inefficient Excel tracking and supporting the Kingdom's Vision 2030 goals.

Chapter 9: References

1. Microsoft. Power BI Technical Documentation (2024).

Chapter 10: Appendices

Include full M-code and detailed Database Schema here.

Appendix A: Power Query M-Code

Table 1:

let

```
Source = Excel.Workbook(File.Contents("C:\Desktop\Chandigarh University Msc\04th Semester\Major Project - Data Analysis using Power BI\Artist Database Dashboard\Sample Data For Artists Database.xlsx"), null, true),
```

```
Table1_Table = Source{[Item="Table1",Kind="Table"]}[Data],
```

```
#"Changed Type" = Table.TransformColumnTypes(Table1_Table,{{"S.N", Int64.Type}, {"Surname/Studio Name/Artistic Name", type text}, {"Name", type text}, {"Nationality", type text}, {"E-Mail Address", type text}, {"Category#(lf)(Artist/Designer/Architect)", type text}, {"Website", type text}, {"Gender", type text}, {"Vetting Process", type text}, {"Birthyear", Int64.Type}, {"Age", Int64.Type}, {"Program", type text}, {"Status Program 1#(lf)(Awarded/Qualified/Not Qualified/Not Awarded)", type text}, {"Artwork", type text}, {"Year", type any}, {"Art Typology ", type text}, {"Location", type text}, {"Notes on Program ", type text}}),
```

```
#"Renamed Columns1" = Table.RenameColumns("#Changed Type",{{"Status Program 1#(lf)(Awarded/Qualified/Not Qualified/Not Awarded)", "Status Program"}})
```

in

```
#"Renamed Columns1"
```

Artwork_Status Table:

let

```
Source = Excel.Workbook(File.Contents("C:\Desktop\Chandigarh University Msc\04th Semester\Major Project - Data Analysis using Power BI\Artist Database Dashboard\Sample Data For Artists Database.xlsx"), null, true),
```

```
Artwork_status_Table = Source{[Item="Artwork_status",Kind="Table"]}[Data],
```

```
#"Changed Type" = Table.TransformColumnTypes(Artwork_status_Table,{{"Sub-Program", type text}, {"Expected Artworks", Int64.Type}, {"Delivered Artworks", Int64.Type}, {"Remaining Artworks", Int64.Type}})
```

in

#"Changed Type"

Appendix B: Detailed Database Schema & Data Types

Table 1:

```
= Table.TransformColumnTypes(Table1_Table,{{"S.N", Int64.Type}, {"Surname/Studio  
Name/Artistic Name", type text}, {"Name", type text}, {"Nationality", type text}, {"E-Mail  
Address", type text}, {"Category#(lf)(Artist/Designer/Architect)", type text}, {"Website",  
type text}, {"Gender", type text}, {"Vetting Process", type text}, {"Birthyear", Int64.Type},  
{"Age", Int64.Type}, {"Program", type text}, {"Status Program  
1#(lf)(Awarded/Qualified/Not Qualified/Not Awarded)", type text}, {"Artwork", type text},  
{"Year", type any}, {"Art Typology ", type text}, {"Location", type text}, {"Notes on Program ",  
type text}})
```

Artwork_Status Table:

```
= Table.TransformColumnTypes(Artwork_status_Table,{{"Sub-Program", type text},  
{"Expected Artworks", Int64.Type}, {"Delivered Artworks", Int64.Type}, {"Remaining  
Artworks", Int64.Type}})
```